

Food Delivery Platform Performance Analysis

(146,200 rows × 29 columns)

1. Project Overview

The **Food Delivery Platform Performance Analysis** project was created to understand how a food delivery business is performing across different business areas such as sales, orders, cities, restaurants, cuisines, delivery operations, promotions, customer experience, weekends, and festivals.

The main purpose of this project is to take raw business data, check its quality, analyze important performance metrics, identify meaningful business patterns, and present the results through an interactive Excel dashboard.

This project follows a practical Data Analyst workflow:

Raw Data → Data Quality Check → Analysis → Dashboard → Business Insights → Recommendations

2. Business Problem

A food delivery platform generates a large amount of operational and sales data every day. However, raw data alone does not tell management where the business is performing well or where improvements are required.

The business needs answers to questions such as:

- How many orders are being generated?
- How much sales and net sales are being generated?
- Is the business growing year over year?
- Which cities contribute the most orders and sales?
- Which restaurants, cuisines, and categories perform better?
- Are deliveries happening earlier, on time, or late?
- Do promotions have a meaningful impact on order value?
- Is customer satisfaction related to ratings?
- How does performance differ between weekdays and weekends?
- How much business happens during festivals?
- What areas should management focus on for future improvement?

This project attempts to answer these questions using the available dataset.

3. Project Objective

The main objective of this project is:

To analyze food delivery platform performance and identify opportunities to improve sales growth, order performance, delivery operations, customer experience, and promotional effectiveness using data-driven insights.

The project also aims to demonstrate the practical use of Excel for a real-world Data Analyst workflow rather than simply performing basic calculations.

4. Dataset Overview

The dataset contains food delivery business information covering the years **2024 and 2025**.

The dataset includes information related to:

- Dates and time periods
- Restaurants
- Cities
- Cuisine types
- Business categories
- Weather
- Weekend and festival indicators
- Promotional activity
- Customer ratings
- Customer satisfaction
- Order values
- Delivery times
- Packaging charges
- Commission
- Sales
- Net sales

The dataset contains a total of **32,099,435 orders** represented in the business-level analysis.

The main cities included in the analysis are:

- Bangalore
- Chennai
- Delhi
- Hyderabad
- Mumbai

- Pune

5. Tools Used

Microsoft Excel

The main analysis was performed using Excel.

The following Excel features were used:

- Excel Tables
- Power Query
- Formulas
- Data Quality Checks
- PivotTables
- PivotCharts
- Dashboard creation
- Data summarization
- Business KPI analysis

6. Data Preparation and Data Quality Checks

Before performing business analysis, the dataset was reviewed for data quality and consistency.

The purpose of this step was to make sure that the data used for analysis was reasonable and internally consistent.

Checks performed included:

- Date range validation
- City validation
- Restaurant-related fields
- Numeric field validation
- Binary field validation
- Delivery time validation
- Delivery delay calculation
- Net Sales consistency
- Commission consistency
- Promotion indicators
- Weekend indicators
- Festival indicators
- Packaging charge indicators
- 24x7 indicators

Important validation results

The dataset covers:

Start Date: 01-Jan-2024

End Date: 31-Dec-2025

The following binary fields were checked:

- **Is_Weekend** → 0 and 1
- **Is_Festival** → 0 and 1
- **Is_Promo** → 0 and 1
- **Has_Packaging_Charges** → 0 and 1
- **Is_24x7** → 0 and 1

The **Is_Peak_Hour** field contained only **0** in the available data. Because there were no records marked as peak hour, this field could not provide a meaningful peak-hour comparison.

Delivery Delay

Delivery delay was calculated as:

Delivery Delay = Delivery Time Actual – Delivery Time Minimum

The observed delay values ranged approximately from **-5 to +5 minutes**.

Negative values represent deliveries that occurred earlier than the expected/minimum delivery time, while positive values represent delays.

Net Sales Validation

Net Sales was checked against the relationship:

Net Sales = Sales – Commission

The overall calculated and recorded values were consistent with the dataset used for the project.

The final recorded totals were:

- **Sales:** ₹9,557,012,837
- **Commission:** ₹1,778,007,160
- **Net Sales:** ₹7,778,562,143

7. Key Business KPIs

The overall business KPIs identified from the analysis were:

KPI	Value
Total Orders	32,099,435
Total Sales	₹9,557,012,837
Total Commission	₹1,778,007,160
Total Net Sales	₹7,778,562,143

These KPIs were used as the main summary metrics in the dashboard.

8. Analysis Performed

The project contains the following major analysis areas:

8.1 Overall Business Performance

Reviewed total orders, sales, commission, and net sales to understand the overall size and performance of the business.

8.2 Year-over-Year Performance

Compared 2024 and 2025 performance to identify changes in orders and sales.

8.3 Monthly Performance

Compared monthly orders and sales between 2024 and 2025 to identify monthly trends and year-over-year changes.

8.4 City Performance

Compared cities based on orders, sales, and net sales to identify high-volume and high-revenue markets.

8.5 Restaurant Performance

Analyzed restaurant-level performance using order and sales metrics.

8.6 Cuisine Performance

Analyzed performance across different cuisine types.

8.7 Category Performance

Compared business performance across different restaurant categories.

8.8 Delivery Performance

Analyzed delivery delay and classified delivery records into:

- Early
- On Time
- Late

8.9 Promotion Performance

Compared promotional and non-promotional business performance using orders, sales, net sales, and value per order.

8.10 Customer Performance

Compared customer ratings, order volume, sales, and average customer satisfaction.

8.11 Weekend Performance

Compared weekday and weekend orders, sales, and net sales.

8.12 Festival Performance

Compared festival and non-festival orders, sales, and net sales.

9. Dashboard

A management-style Excel dashboard was created to provide a quick summary of the business.

KPI Cards

The dashboard contains:

- Total Orders
- Total Sales
- Total Commission
- Total Net Sales

Dashboard Visuals

The dashboard contains the following charts:

1. **Monthly Sales Trend — 2024 vs 2025**
2. **Sales by City**
3. **Delivery Performance**
4. **Sales: Promo vs Non-Promo**
5. **Orders by Customer Rating**

The purpose of the dashboard is to allow a manager or decision-maker to quickly understand the overall business performance without going through every PivotTable.

10. Key Business Findings

10.1 Year-over-Year Growth

The business showed positive growth from 2024 to 2025.

- Orders increased by approximately **7.04%**
- Sales increased by approximately **7.06%**

This indicates that the platform experienced overall growth in both order volume and sales.

10.2 City Performance

Mumbai generated the highest sales, with approximately **₹1.68 billion** in sales.

Bangalore recorded the highest order volume, with approximately **6.72 million orders**.

This shows that order volume and revenue contribution are not exactly the same across cities.

10.3 Delivery Performance

The average delivery delay was approximately **0 minutes**.

The observed delay values ranged from approximately **-5 to +5 minutes**.

This indicates that actual delivery times generally stayed close to the expected delivery time in the available data.

10.4 Promotion Performance

The average Sales per Order was:

- **Non-Promo: ₹297.76**
- **Promo: ₹297.55**

The average Net Sales per Order was:

- **Non-Promo:** ₹242.34
- **Promo:** ₹242.21

The difference is very small.

Therefore, the available data does not show a meaningful difference in average order value between promotional and non-promotional orders.

However, this does **not** prove that promotions are ineffective because the dataset does not contain enough information to calculate the complete cost and incremental return of each promotion.

10.5 Customer Performance

The records with a **4.7 rating** had:

- Orders: 1,176,537
- Sales: ₹320,558,708
- Average Customer Satisfaction: 0.610

The records with a **3.8 rating** had:

- Orders: 1,405,699
- Sales: ₹408,902,680
- Average Customer Satisfaction: 0.556

Higher-rated records showed higher average customer satisfaction, but they did not automatically generate higher orders or sales.

Therefore, customer rating and sales should not be treated as directly causing each other based on this analysis alone.

10.6 Weekend Performance

Weekdays generated:

- **23,005,799 orders**
- **₹6,849,528,311 sales**

Weekends generated:

- **9,093,636 orders**
- **₹2,707,484,526 sales**

Weekend activity represented approximately **28.3% of total orders and sales**.

The majority of business activity therefore occurred on weekdays.

10.7 Festival Performance

Non-festival periods generated:

- **30,709,104 orders**
- **₹9,143,082,625 sales**

Festival periods generated:

- **1,390,331 orders**
- **₹413,930,212 sales**

Festival activity represented approximately **4.3% of total orders and sales**.

This means festivals contributed a smaller portion of the overall business volume in this dataset.

11. Business Recommendations

Based on the analysis, the following recommendations can be considered:

11.1 Focus on High-Performing Cities

Mumbai generated the highest sales while Bangalore generated the highest order volume.

The business can use different strategies for different cities instead of applying the same strategy everywhere.

11.2 Maintain Delivery Reliability

The average delivery delay was close to zero.

The business should continue monitoring delivery performance, especially as order volume increases, to prevent late deliveries from increasing.

11.3 Evaluate Promotions Based on Actual Business Impact

Promotional and non-promotional orders showed almost identical sales and net sales per order.

Future promotion analysis should consider:

- Discount amount
- Promotion cost
- Additional orders generated
- New customers acquired

- Customer retention
- Incremental revenue

This would provide a better measure of promotion effectiveness.

11.4 Improve Customer Experience

Higher-rated records showed higher customer satisfaction.

The business should continue improving the customer experience while also considering other factors such as price, restaurant quality, delivery experience, cuisine, and location.

11.5 Use Targeted Weekend Campaigns

Since weekdays currently contribute the majority of business volume, weekends can be used for targeted campaigns and offers designed specifically to increase weekend demand.

11.6 Use Festivals as Targeted Opportunities

Festival periods currently contribute a smaller share of total business.

Rather than depending heavily on festivals, the business can use specific festivals for targeted campaigns and measure the additional business generated by each campaign.

12. Project Limitations

The analysis also has some limitations because of the information available in the dataset.

Promotion ROI

The dataset does not provide complete promotion or discount costs.

Therefore, true promotion ROI cannot be calculated.

Profitability

Net Sales was calculated as:

Sales – Commission

Therefore, Net Sales should not be interpreted as final company profit because other business expenses and operating costs are not available.

Customer-Level Analysis

The available data does not provide enough customer-level information to perform detailed customer retention, churn, lifetime value, or repeat-purchase analysis.

Causation

The analysis identifies patterns and relationships in the available data.

It does not prove that a specific factor directly caused an increase or decrease in sales.

Peak Hour Analysis

The `Is_Peak_Hour` field contained only 0 values in the available dataset.

Because there were no records marked as peak hour, a meaningful peak-hour versus non-peak-hour comparison could not be performed.

13. Overall Conclusion

This project demonstrates how raw food delivery business data can be transformed into useful business information using Excel.

The analysis showed positive year-over-year growth, differences in city-level performance, stable delivery timing, similar order values between promotional and non-promotional orders, differences in customer satisfaction across ratings, and different business patterns across weekdays, weekends, and festivals.

The main purpose of the project was not only to calculate numbers but to understand **what those numbers mean for the business**.

The final dashboard provides management with a quick view of important KPIs and performance trends, while the insights and recommendations provide possible areas for further investigation and improvement.

Overall, this project demonstrates a practical Data Analyst workflow:

Data → Data Quality → Analysis → Visualization → Insights → Recommendations

14. Skills Demonstrated

This project demonstrates practical experience with:

- Microsoft Excel
- Power Query
- Data Cleaning and Validation
- Data Quality Checks

- PivotTables
- PivotCharts
- KPI Analysis
- Trend Analysis
- Year-over-Year Analysis
- Business Performance Analysis
- Data Visualization
- Dashboard Development
- Business Insights
- Data-Driven Recommendations
- Analytical Thinking
- Business Problem Solving